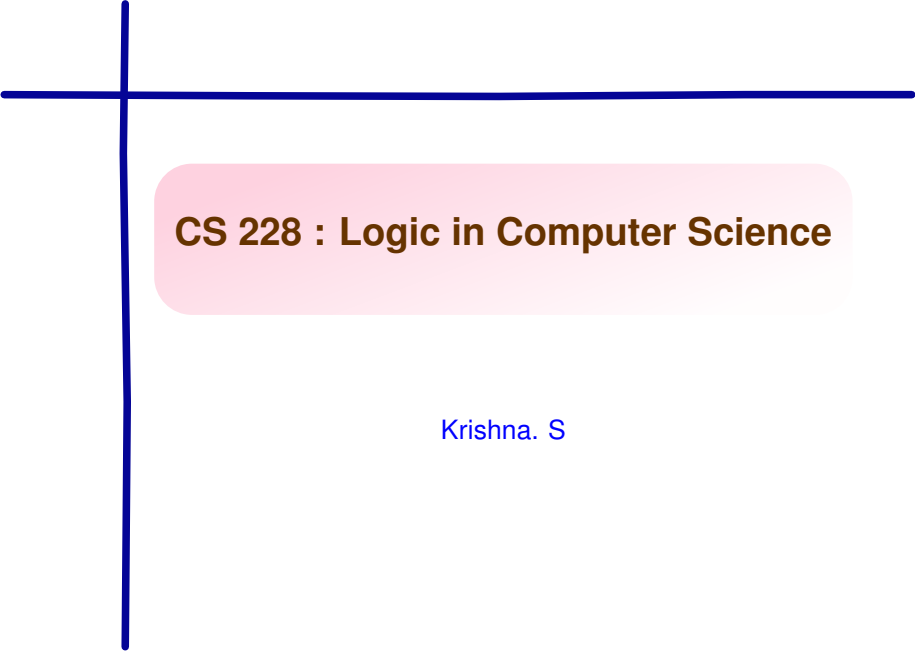


A decorative blue crosshair consisting of a vertical line and a horizontal line intersecting in the upper-left quadrant of the slide.

CS 228 : Logic in Computer Science

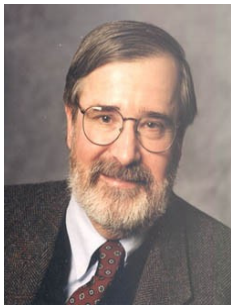
Krishna. S

We know

- ▶ Model checking when the property is FO-definable, and the system is abstracted as a finite automaton
- ▶ Model checking when the property is MSO-definable, and the system is abstracted as a finite automaton
- ▶ System behaviours are finite words

Linear Temporal Logic

Model Checking

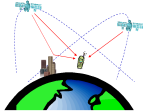


- ▶ Year 2007 : ACM confers the **Turing Award** to the pioneers of Model Checking: **Ed Clarke, Allen Emerson, and Joseph Sifakis**
- ▶ https://amturing.acm.org/award_winners/clarke_1167964.cfm

Model checking

- ▶ Model checking has evolved in last 25 years into a widely used verification and debugging technique for software and hardware.
- ▶ Model checking used (and further developed) by companies/institutes such as IBM, Intel, NASA, Cadence, Microsoft, and Siemens, and has culminated in many freely downloadable software tools that allow automated verification.

What is Model Checking?



System

satisfy?



specification

good/bad properties

What is Model Checking?



System

satisfy?



specification

System Model

model-checking

$\models?$

Spec Model

ϕ

Model Checker as a Black Box

- ▶ Inputs to Model checker : A finite state system M , and a property P to be checked.
- ▶ Question : Does M satisfy P ?
- ▶ Possible Outputs
 - ▶ Yes, M satisfies P
 - ▶ No, here is a counter example!.

What are Models?

Transition Systems

- ▶ States labeled with propositions
- ▶ Transition relation between states
- ▶ Action-labeled transitions to facilitate composition

What are Properties?

Example properties

- ▶ Can the system reach a deadlock?
- ▶ Can two processes ever be together in a critical section?
- ▶ On termination, does a program provide correct output?

Notations for Infinite Words

- ▶ Σ is a finite alphabet
- ▶ Σ^* set of finite words over Σ
- ▶ An infinite word is written as $\alpha = \alpha(0)\alpha(1)\alpha(2)\dots$, where $\alpha(i) \in \Sigma$
- ▶ Such words are called ω -words
- ▶ $a^\omega, a^7.b^\omega$

Transition Systems

A **Transition System** is a tuple $(S, Act, \rightarrow, I, AP, L)$ where

- ▶ S is a set of **states**
- ▶ Act is a set of **actions**
- ▶ $s \xrightarrow{\alpha} s'$ in $S \times Act \times S$ is the **transition relation**
- ▶ $I \subseteq S$ is the **set of initial states**
- ▶ AP is the set of **atomic propositions**
- ▶ $L : S \rightarrow 2^{AP}$ is the **labeling function**

Traces of Transition Systems

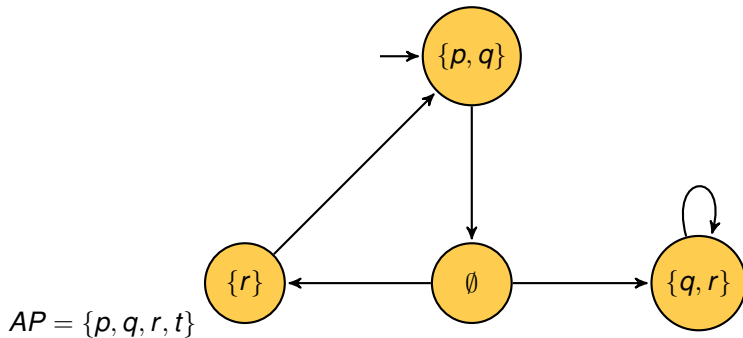
- ▶ Labels of the locations represent values of all observable propositions $\in AP$
- ▶ Captures system state
- ▶ Focus on sequences $L(s_0)L(s_1) \dots$ of labels of locations
- ▶ Such sequences are called **traces**
- ▶ Assuming transition systems have no terminal states,
 - ▶ Traces are infinite words over 2^{AP}
 - ▶ Traces $\in (2^{AP})^\omega$
 - ▶ Go to the example slide and define traces

Traces of Transition Systems

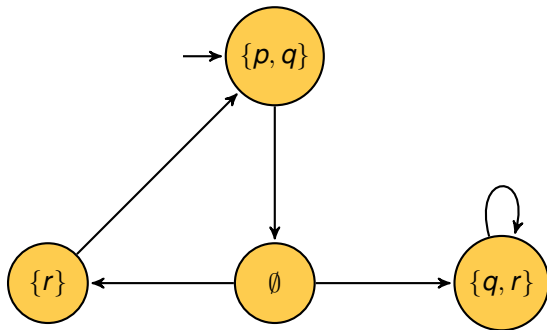
Given a transition system $TS = (S, Act, \rightarrow, I, AP, L)$ without terminal states,

- ▶ All maximal executions/paths are infinite
- ▶ Path $\pi = s_0 s_1 s_2 \dots$, $trace(\pi) = L(s_0)L(s_1)\dots$
- ▶ For a set Π of paths, $Trace(\Pi) = \{trace(\pi) \mid \pi \in \Pi\}$
- ▶ For a location s , $Traces(s) = Trace(Paths(s))$
- ▶ $Traces(TS) = \bigcup_{s \in I} Traces(s)$

Example Traces



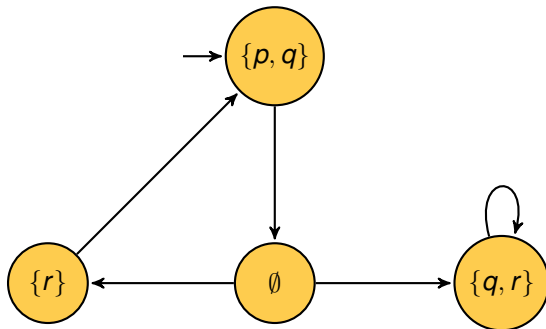
Example Traces



$AP = \{p, q, r, t\}$

► $\{p, q\} \emptyset \{q, r\}^\omega$

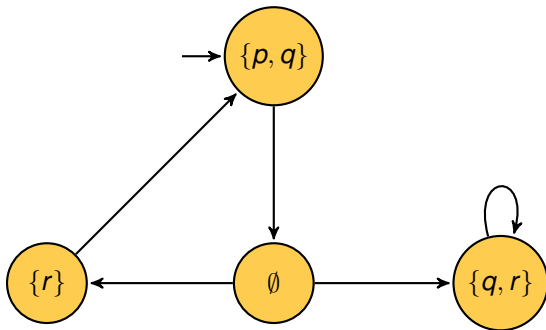
Example Traces



$AP = \{p, q, r, t\}$

- ▶ $\{p, q\} \emptyset \{q, r\}^\omega$
- ▶ $(\{p, q\} \emptyset \{r\})^\omega$

Example Traces



$AP = \{p, q, r, t\}$

- ▶ $\{p, q\} \emptyset \{q, r\}^\omega$
- ▶ $(\{p, q\} \emptyset \{r\})^\omega$
- ▶ $(\{p, q\} \emptyset \{r\})^* \{p, q\} \emptyset \{q, r\}^\omega$

Linear Time Properties

- ▶ Linear-time properties specify traces that a TS must have
- ▶ A LT property P over AP is a subset of $(2^{AP})^\omega$
- ▶ TS over AP satisfies a LT property P over AP

$$TS \models P \text{ iff } \text{Traces}(TS) \subseteq P$$

- ▶ $s \in S$ satisfies LT property P (denoted $s \models P$) iff $\text{Traces}(s) \subseteq P$